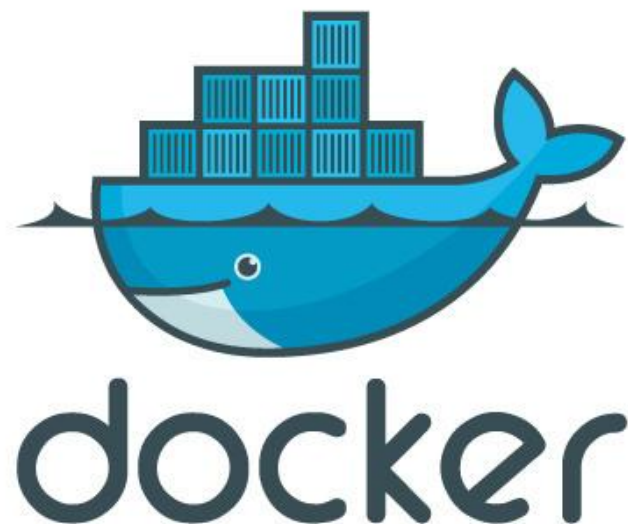




<https://k8s-school.fr>



# Conteneurs: l'exemple de Docker

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# Plan

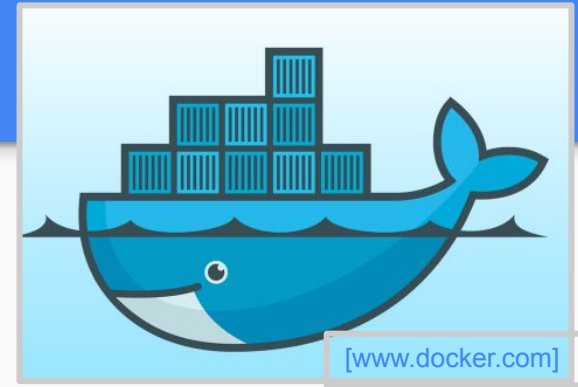
- What is Docker?
  - Docker vs. Virtual Machine
  - History, Status, Run Platforms
  - Hello World
- Images and Containers
- Volume Mounting, Port Publishing, Linking
- Around Docker, Docker Use Cases
- Hands-On Workshop

# Qu'est ce que Docker?

*Docker is an open-source project that automates the deployment of applications inside software containers, by providing an additional layer of abstraction and automation of operating system–level virtualization on Linux.*

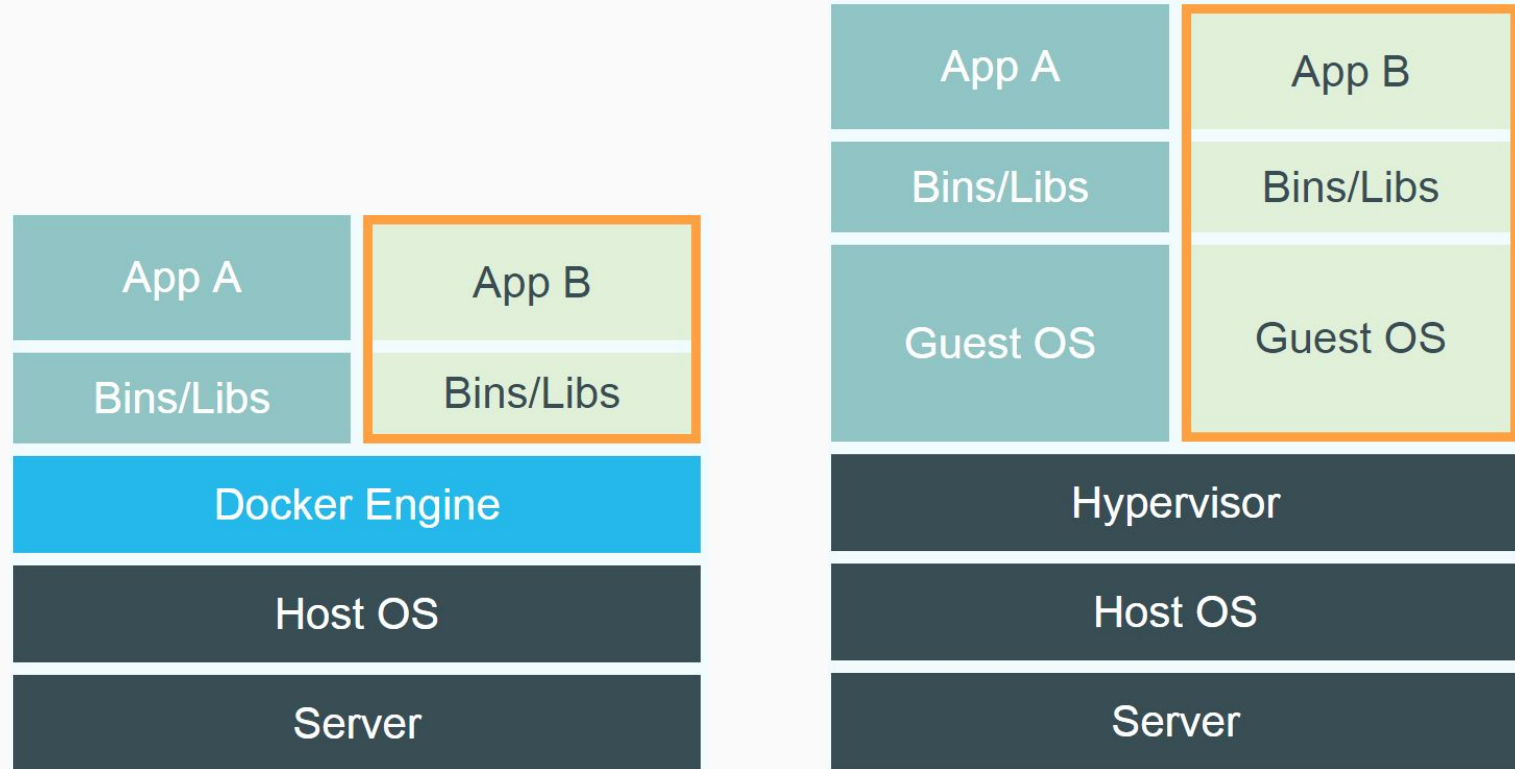
[Source: [en.wikipedia.org](https://en.wikipedia.org)]

# Docker: Name



- Provide a uniformed wrapper around a software package: «*Build, Ship and Run Any App, Anywhere*»  
[www.docker.com]
  - Similar to shipping containers:  
The container is always the same, regardless of the contents and thus fits on all trucks, cranes, ships, ...

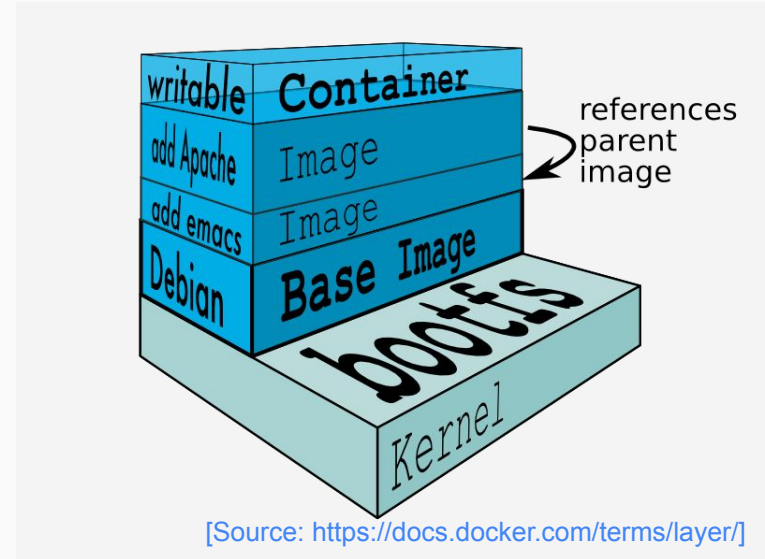
# Docker vs. Virtual Machine



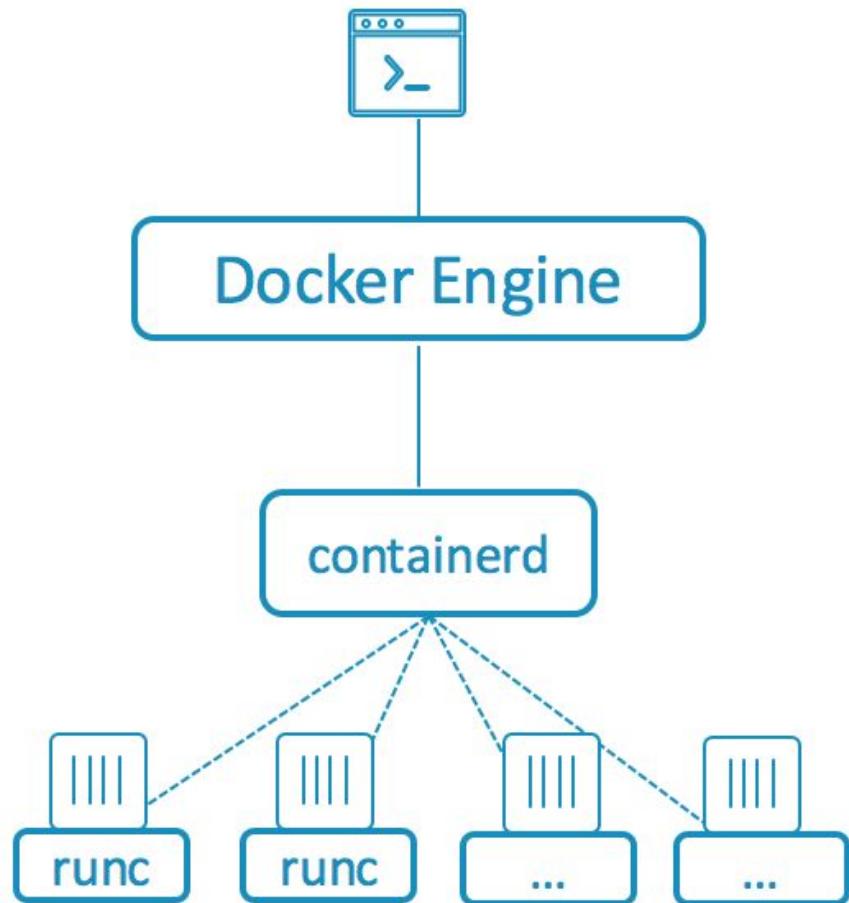
Source: <https://www.docker.com/whatisdocker/>

# Docker Technology

- runC: The universal container runtime:  
<https://runc.io/>
- Open Containers Initiative:  
<https://www.opencontainers.org/>
- Layered File System  
<https://docs.docker.com/engine/userguide/storagedriver/imagesandcontainers/>



# Modular architecture



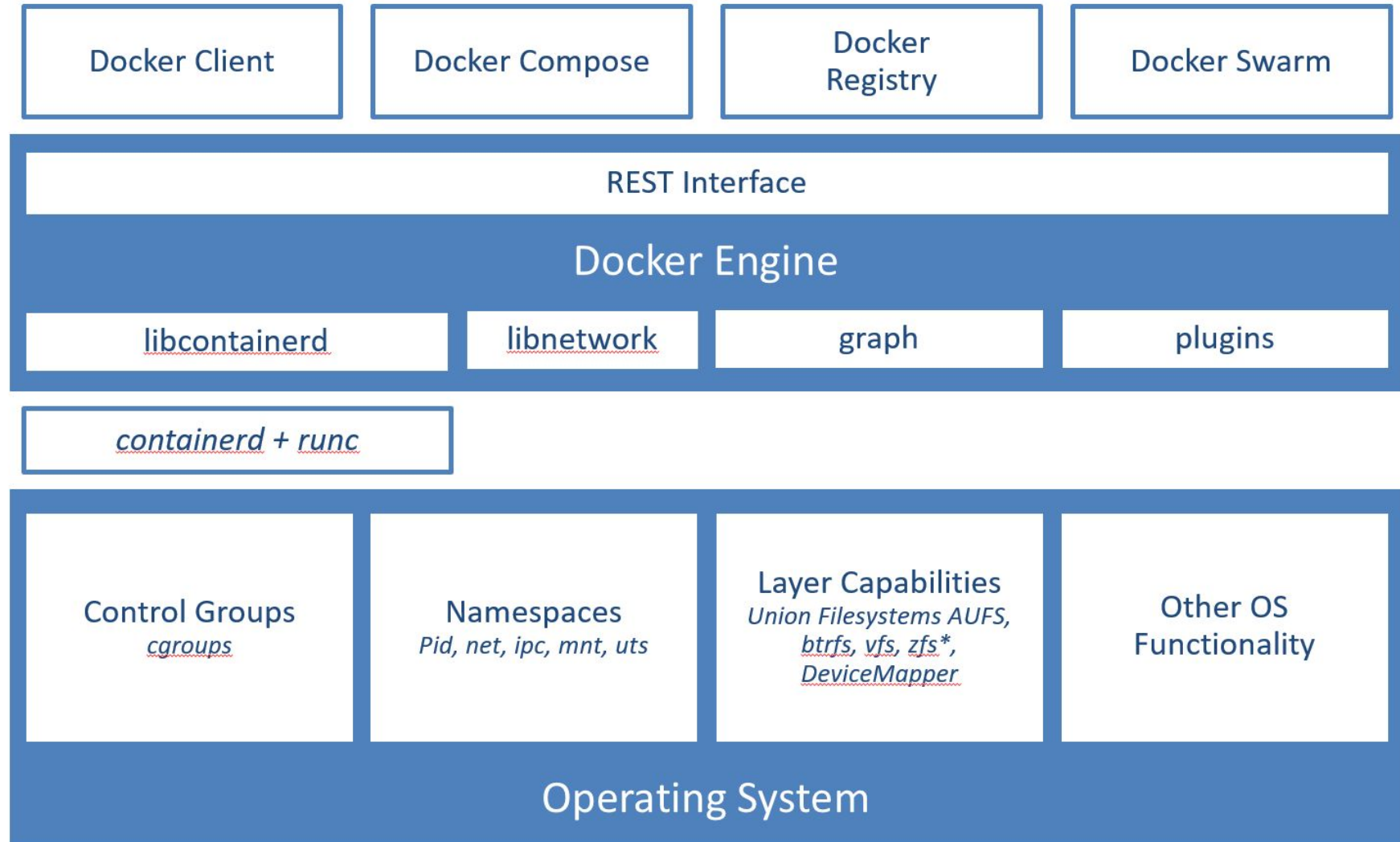
Same Docker UI and commands

User interacts with the Docker Engine

Engine communicates with containerd

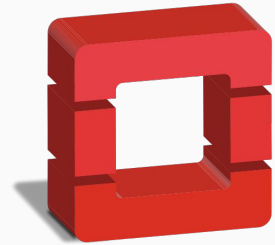
containerd spins up runc or other OCI compliant runtime to run containers

# Architecture in Linux



# Run Platforms

- Various Linux distributions (Ubuntu, Fedora, RHEL, Centos, openSUSE, ...)
- Cloud (Amazon EC2, Google Compute Engine, Microsoft Azure)



**openstack.**  
CLOUD SOFTWARE



Microsoft Azure



**Core** OS

# Hello world

## Simple Command - Ad-Hoc Container

- `docker run ubuntu echo  
Hello World`
  - `docker images [-a]`
  - `docker ps -a`

# Terminology - Image

- Persisted snapshot that can be run
  - *images*: List all local images
  - *run*: Create a container from an image and execute a command in it
  - *tag*: Tag an image
  - *pull*: Download image from repository
  - *rmi*: Delete a local image
    - This will also remove intermediate images if no longer used

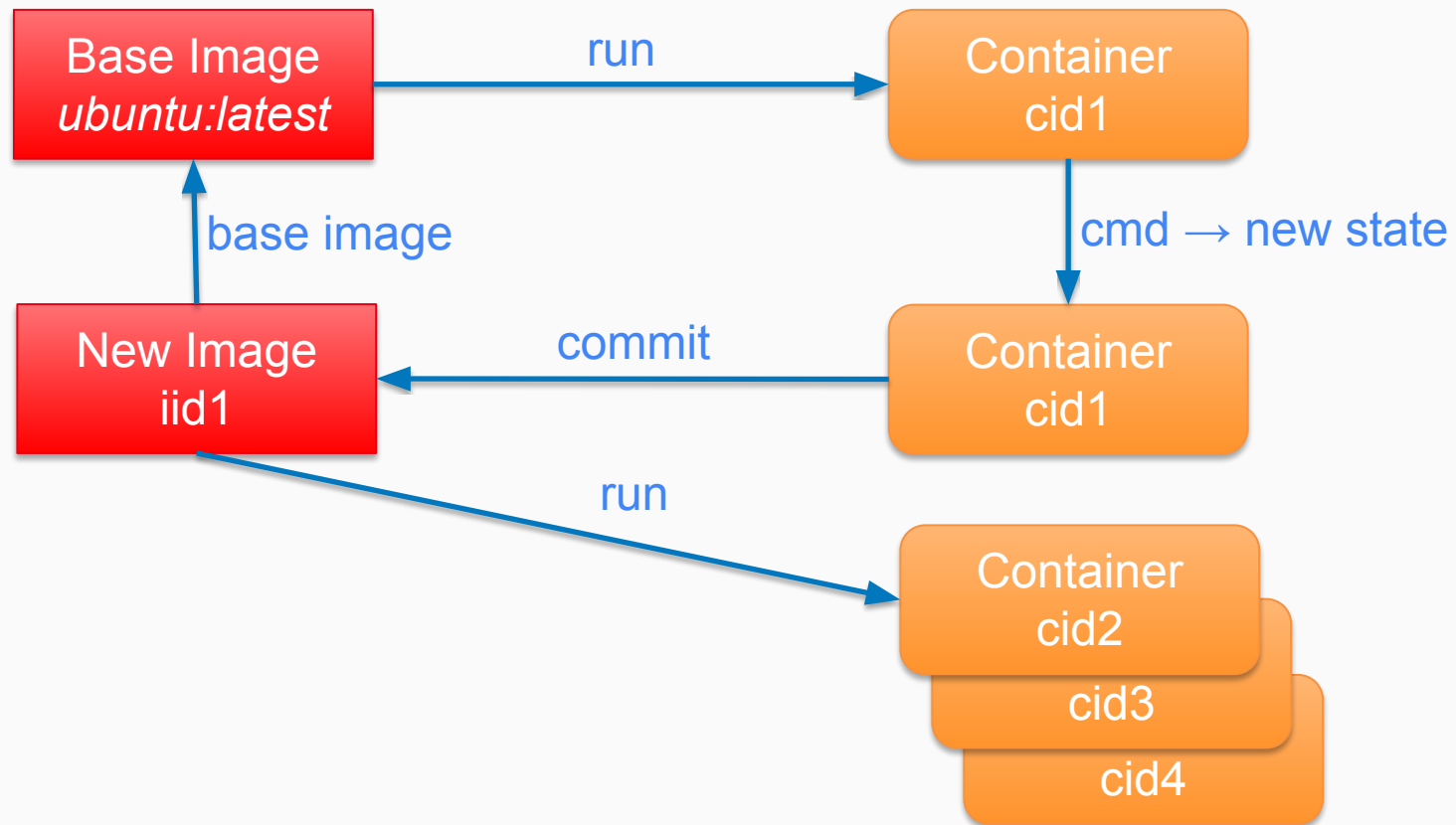
# Terminology - Container

Runnable instance of an image

- *ps*: List all running containers
- *ps -a*: List all containers (incl. stopped)
- *top*: Display processes of a container
- *start*: Start a stopped container
- *stop*: Stop a running container
- *pause*: Pause all processes within a container
- *rm*: Delete a container
- *commit*: Create an image from a container



# Image vs. Container



# Dockerfile

- Create images automatically using a build script: «Dockerfile»
- Can be versioned in a version control system like Git or SVN, along with all dependencies
- Docker Hub can automatically build images based on dockerfiles on Github

# Dockerfile Example

- Dockerfile:

```
FROM ubuntu
```

```
ENV DOCK_MESSAGE Hello My World
```

```
ADD dir /files
```

```
CMD ["bash", "/files/some-script.sh"]
```

- `docker build [DockerFileDir]`
- `docker inspect [imageId]`

# Mount Volumes

- `docker run -ti -v /hostLog:/log ubuntu`

- *Run second container: Volume can be shared*

```
docker run -ti --volumes-from  
1stContainerName ubuntu
```

# Publish Port

- `docker run -t -p 8080:80 ubuntu netcat -l -p 80`
  - Map container port 80 to host port 8080
  - Check on host: `netcat localhost 8080`
- *Link with other docker container*
  - `docker run -ti --link containerName:alias ubuntu`
  - *See link info with `set`*

# Around Docker

- Docker Images: Docker Hub
- Vagrant: «Docker for VMs»
- Automated Setup
  - Puppet, Chef, Ansible, ...
- Docker Ecosystem
  - kubernetes, swarm, nomad
  - compose

# Docker Hub

- Public repository of Docker images
  - <https://hub.docker.com/>
  - docker search [term]
- Automated: Has been automatically built from Dockerfile
  - Source for build is available on GitHub

# Resource Usage

- `docker top [containerId]`
- `docker free -m [containerId]`
- `docker ps`
- Start 100 WebServer containers  
`docker run -d -p $hostPort:5000 training/webapp`

# Docker Use Cases

- Development Environment
- Environments for Integration Tests
- Quick evaluation of software
- Microservices
- Multi-Tenancy
- Unified execution environment:

dev → test → prod (local, VM, cloud, ...)